WORKED SOLUTION

STEP I 2004, Question 5

Pure • Residues Modulo Five

STEP I 2004, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/04-s1-q5

The key idea

The five progressions sort the positive integers by their remainder on division by 5. Once each class has a general term like $5n-3$, sums and squares can be tracked by expanding and pulling out a factor of 5. That gives a complete picture of where squares can land, and both impossibility proofs follow from it: the right-hand side lands in a class that the left-hand side can never reach.

Solution

The general terms

Each progression has common difference 5, so for $n = 1, 2, 3, \ldots$ the nth terms are

$$A:\ 5n-4, \qquad B:\ 5n-3, \qquad C:\ 5n-2, \qquad D:\ 5n-1, \qquad E:\ 5n.$$

(Put $n = 1$ to check that each gives the first listed term.) So a positive integer is in A, B, C, D or E exactly when it leaves remainder 1, 2, 3, 4 or 0 on division by 5.

Sum of a term in B and a term in C

Take any term $5m-3$ of B and any term $5n-2$ of C, where $m, n \geq 1$. Then

$$(5m-3)+(5n-2) = 5m+5n-5 = 5(m+n-1).$$

Since $m+n-1 \geq 1$ is a positive integer, this is the $(m+n-1)$th term of E, as required.

Squares of terms in B and C

For a term $5m-3$ of B,

$$\begin{aligned}(5m-3)^2 &= 25m^2-30m+9\\ &= 5(5m^2-6m+2)-1.\end{aligned}$$

Here $k = 5m^2 - 6m + 2$ is an integer, and $k = 5m^2 - 6m + 2 = m(5m - 6) + 2 \geq 1$ for every $m \geq 1$ (at $m = 1$ it is 1, and for $m \geq 2$ both m and $5m - 6$ are positive). So $(5m - 3)^2 = 5k - 1$ is the kth term of D. Every square of a term in B is in D.

The similar claim for C: *the square of every term in C is also a term in D.* Indeed, for a term $5m - 2$ of C,

$$\begin{aligned}(5m - 2)^2 &= 25m^2 - 20m + 4 \\ &= 5(5m^2 - 4m + 1) - 1,\end{aligned}$$

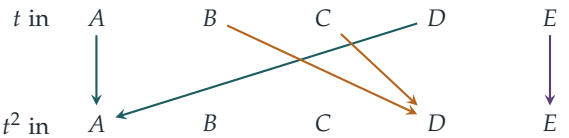
and $k = 5m^2 - 4m + 1 = m(5m - 4) + 1 \geq 2$ for $m \geq 1$. So $(5m - 2)^2$ is the kth term of D.

? **Ask yourself** Where do the squares of terms in A, D and E go?

We need this for parts (i) and (ii), so it is worth doing the other three classes in the same way:

$$\begin{aligned}(5m - 4)^2 &= 25m^2 - 40m + 16 = 5(5m^2 - 8m + 4) - 4 && \text{(in } A\text{)}, \\ (5m - 1)^2 &= 25m^2 - 10m + 1 = 5(5m^2 - 2m + 1) - 4 && \text{(in } A\text{)}, \\ (5m)^2 &= 5(5m^2) && \text{(in } E\text{)}.\end{aligned}$$

The brackets are positive integers for $m \geq 1$ (for instance $5m^2 - 8m + 4 = m(5m - 8) + 4 \geq 1$, and $5m^2 - 2m + 1 \geq 4$). So squaring sends the classes as follows:



In particular, *no perfect square lies in B or C*: every positive square lies in A, D or E.

Part (i)

Suppose that positive integers x and y satisfy $x^2 + 5y = 243\,723$. Since $243\,723 = 5 \times 48\,744 + 3$, the right-hand side is the term $5 \times 48\,745 - 2$ of C. Then

$$x^2 = 243\,723 - 5y = 5(48\,745 - y) - 2.$$

This is positive (it is x^2), so $48\,745 - y$ is a positive integer, and x^2 is a term of C. But we showed that no square lies in C. This contradiction shows that there are no such positive integers x and y.

Part (ii)

A fourth power is the square of a square. The square x^2 lies in A, D or E, and squaring a term of A or D gives a term of A, while squaring a term of E gives a term of E. So for

any positive integer x,

$$x^4 \text{ is in } A \text{ or } E, \quad \text{that is,} \quad x^4 \text{ leaves remainder 1 or 0 on division by 5.}$$

The same holds for y^4, so $2y^4$ leaves remainder 2 or 0. Adding, the remainder of $x^4 + 2y^4$ on division by 5 is the remainder of one of

$$1+2=3, \qquad 1+0=1, \qquad 0+2=2, \qquad 0+0=0,$$

so $x^4 + 2y^4$ is in C, A, B or E, and *never* in D.

Ask yourself Why can we just add the remainders?

If $x^4 = 5a + r$ and $2y^4 = 5b + s$, then $x^4 + 2y^4 = 5(a+b) + (r+s)$, so the sum differs from $r+s$ by a multiple of 5 and is in the same class as $r+s$. This is exactly the calculation we did for $B + C$ above, in general form.

Now $26\,081\,974 = 5 \times 5\,216\,395 - 1$ ends in 4, so it is a term of D. Hence it cannot equal $x^4 + 2y^4$ for any positive integers x and y, as required.

Checks

- Small cases: $2^2 = 4$, $7^2 = 49$ and $12^2 = 144$ all end in 4 or 9, so lie in D; $3^2 = 9$, $8^2 = 64$, $13^2 = 169$ likewise.
- Fourth powers: $1, 16, 81, 256, 625, 1296$ all end in 1, 6 or 5, so lie in A or E.
- A computer search over all $x \le \sqrt{243\,723} \approx 493.7$ finds no x with $243\,723 - x^2$ a positive multiple of 5, and a search over all x with $x^4 < 26\,081\,974$ (so $x \le 71$) finds no y with $2y^4 = 26\,081\,974 - x^4$.

Common slips

- Writing the general term of E as $5n+5$ and of A as $5n+1$ is fine, but then n starts at 0; mixing the two conventions gives terms that are off by one.
- In part (i), forgetting to use that $x^2 > 0$ so $48\,745 - y$ is a *positive* integer; without this, x^2 need not be a term of C as the progressions only contain positive integers.
- In part (ii), handling only x^4 and forgetting that the factor 2 moves y^4 from remainder 1 to remainder 2.

Beyond STEP

The five progressions are the residue classes modulo 5, and the whole question is arithmetic in $\mathbb{Z}/5\mathbb{Z}$: the squares there are $0, 1, 4$ (the *quadratic residues*) and the fourth powers are $0, 1$, an instance of Fermat's little theorem $a^4 \equiv 1 \pmod 5$ for $5 \nmid a$. Reducing a Diophantine equation modulo a well-chosen number is the first tool for proving it has

no solutions. Davenport's *The Higher Arithmetic* develops congruences and quadratic residues beautifully.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.